Fourier Neural Operator for Parametric Darcy Flow

AE646: Scientific Machine Learning for Fluid Mechanics
Project Theme #8 - Operator Learning for Parametric PDEs

Team PINNacles

Aryamann Srivastava • Varun Sathaye • Atishay Jain • Vedant S. Tiwari

IIT Kanpur | Semester 2026–27-I | Dr. Sathesh Mariappan

Problem Statement

2D Darcy flow - steady-state pressure in a heterogeneous porous medium

The Governing Equation

$$-\nabla\cdot(\,\kappa(x,y)\,\nabla u(x,y)\,) = f(x,y) \quad \text{on } (0,1)^2$$

$u = 0$ on boundary, $f = \beta = 1.0$

$\kappa(x,y)$ = permeability field | $u(x,y)$ = pressure solution

The Challenge

- Traditional solvers (FDM/FEM) re-solve for every new κ field
- Each solve: ~1 second for a simple 2D case
- Optimization or UQ needs thousands of solves → impractical

Objective

Learn the solution operator $G : \kappa(x,y) \rightarrow u(x,y)$ once.

Evaluate any new permeability field in milliseconds - no re-solving needed.

Enables fast parametric studies, optimization, and uncertainty quantification.

Dataset

PDEBench 2D Darcy Flow (β=1.0) - official, checksum-verified source

Official Source

DaRUS / University of Stuttgart
DOI: 10.18419/darus-2986
10,000 samples at native 128×128 resolution

Our Reproducible Subset (seed = 42)

Train: 900 samples
Validation: 100 samples
Test: 200 samples (held out)
Main resolution: 64×64 (stride-2 downsample)

Permeability κ(x,y)

Piecewise-constant: values in {0.1, 1.0}

Generated by thresholding a smooth Gaussian random field at zero

Produces sharp-boundary blob patterns in 2D

Why 64×64?

Standard resolution for FNO Darcy experiments

Retains full PDE spatial structure

FNO can then be tested at 128×128 zero-shot - no retraining needed

Super-Res Hold-out

200 test samples also kept at native 128×128

Used only for zero-shot evaluation

Never seen by the model during training in any form

Baseline Method: MLP

Dense fully-connected network - no spatial inductive bias

Input 64×64×3 → Flatten →12,288 → Linear 2048 +GELU → Linear 2048 +GELU → Linear 2048 +GELU → Linear 4096 +Reshape → Output 64×64×1

Architecture Details

- 42.0M trainable parameters
- Input: [permeability, x_coord, y_coord] - 3 channels
- 3 hidden layers × 2048 units, GELU activation
- Output: flat 64×64, reshaped to grid

Why This as Baseline?

- Treats all 4096 output pixels independently
- No spatial locality or translation equivariance
- Fixed 64×64 I/O - cannot evaluate at any other resolution
- Establishes a clear performance floor for comparison

Key limitation: the MLP has hard-coded 64×64 input/output dimensions - it cannot be evaluated zero-shot at any other resolution. This is in direct contrast to the FNO.

Proposed SciML Method: Fourier Neural Operator

Operator learning via spectral convolutions - resolution-invariant by construction

Input 64×64×3	→	Lift 3→64	→	SpectralConv Block ×4	→	FFT→W→IFFT +Conv1×1 +LN+GELU	→	Project 64→128→1	→	Output 64×64×1

How Spectral Convolution Works

1. Apply 2D FFT → frequency domain representation
2. Keep only the lowest 12×12 Fourier modes
3. Multiply by learned complex weights (learning in Fourier space)
4. Inverse FFT → back to physical space
5. Add 1×1 Conv bypass, normalize, activate (GELU)

Key Properties

- 4.7M parameters - 8.8× fewer than MLP
- Resolution-invariant: weights don't depend on grid size
    - → Evaluates at 128×128 zero-shot, no retraining
- Spectral inductive bias: Darcy pressure is low-frequency dominated
- Also training a larger config:
    - → width=128, modes=20, 6 layers - 78.8M params
    - → Studies accuracy vs parameter-count trade-off

Evaluation Plan

Metrics, expected results, and planned figures

Primary Metric - Relative L2 Error (physical units)

$\| \text{prediction} - \text{truth} \|_2 / \| \text{truth} \|_2$ computed after denormalizing to physical pressure units

Relative L2 Error

Mean / median / std / min / max across all 200 test samples

Zero-Shot Super-Res

FNO at native 128×128 vs real PDEBench ground truth - no retraining

Inference Speed

Wall-clock time vs FDM numerical solver - measured, not asserted

Qualitative Plots

κ input | target u | prediction | absolute error maps

Literature Reference & Expected Range

Li et al. 2021 (original FNO paper) reports ~0.01–0.02 relative L2 on Darcy flow. Their dataset uses smooth, continuous log-permeability ($\kappa = \exp(\text{GRF})$). PDEBench uses piecewise-constant $\kappa \in \{0.1, 1.0\}$ - sharper interfaces, a harder variant. Some deviation from the literature number is expected and will be discussed.

Expected Figures & Tables

Planned deliverables for Stage 2 and Stage 3 reports

1. Error Histograms

Per-sample relative L2 for FNO original, FNO improved, and MLP baseline.
Reveals mean, median, and tail behaviour.

[Figure]

2. Sample Predictions

Side-by-side: κ input, target u, FNO prediction, absolute error.
Shown for several representative test cases.

[Figure]

3. Model Comparison Table

Rel L2 (mean/ median), parameter count, and inference time for all three models in one table.

[Figure]

4. Zero-Shot Super-Res

FNO evaluated at native 128x128 vs PDEBench ground truth.
No retraining - validates resolution-

[Figure]

5. Efficiency Scatter

Rel L2 vs parameter count - one point per model.
Shows the accuracy vs model-size trade-off.

[Figure]

Work Plan

Three-stage project timeline - Stage 1 complete

Stage 1 ✓ DONE

Project Proposal

Due: Aug 27, 2026

- ✓ Confirm PDEBench as dataset (source verified)
- ✓ Finalize MLP and FNO architecture design
- ✓ Define evaluation methodology and metrics
- ✓ Submit proposal PDF and presentation

Stage 2 UPCOMING

Implementation & Interim Results

Due: Sep 8, 2026

- → Set up data download and preprocessing pipeline
- → Implement and train MLP baseline
- → Implement and train FNO (original config)
- → Gather preliminary results and plots

Stage 3 PLANNED

Final Report, Code & Viva

Due: Nov 13, 2026

- → Train improved FNO (width=128, modes=20)
- → Zero-shot super-resolution at 128×128
- → Inference speed benchmark vs FDM solver
- → Final report, code submission, viva

Individual Contributions

Team PINNacles - Group of 4

Aryamann Srivastava

- Model implementation (FNO and MLP)
- Training and evaluation pipelines
- Data download and preprocessing
- Benchmarks and report drafting

Varun Sathaye

- Literature review
- Baseline model tuning
- Error analysis
- Formatting deliverables

Atishay Jain

- Data visualization and plotting
- Error distribution figures
- Field comparison plots
- Slide preparation

Vedant S. Tiwari

- Zero-shot super-res testing
- Speed benchmarking
- Cross-checking results
- Proofreading final report

AI Tool Use Declaration

Claude / ChatGPT used for conceptual understanding, coding assistance, debugging, and report drafting. All output inspected, verified, and fully understood by the team before submission - in accordance with AE646's academic integrity policy.

Summary

Problem	2D Darcy flow - learn G: $\kappa \rightarrow u$ to replace repeated PDE solves
Dataset	Real PDEBench 2D Darcy (β=1.0) | 900 / 100 / 200 split at 64×64
Baseline	MLP (42M params) - no spatial inductive bias, fixed resolution
SciML	FNO (4.7M params) - spectral convolutions, resolution-invariant
Extended	Larger FNO (78.8M params) - accuracy vs parameter-count trade-off
Evaluation	Rel L2 error | zero-shot super-res at 128×128 | speed vs FDM

Thank you - Questions welcome